

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Q1. Cloud Architecture for a Video Streaming Platform

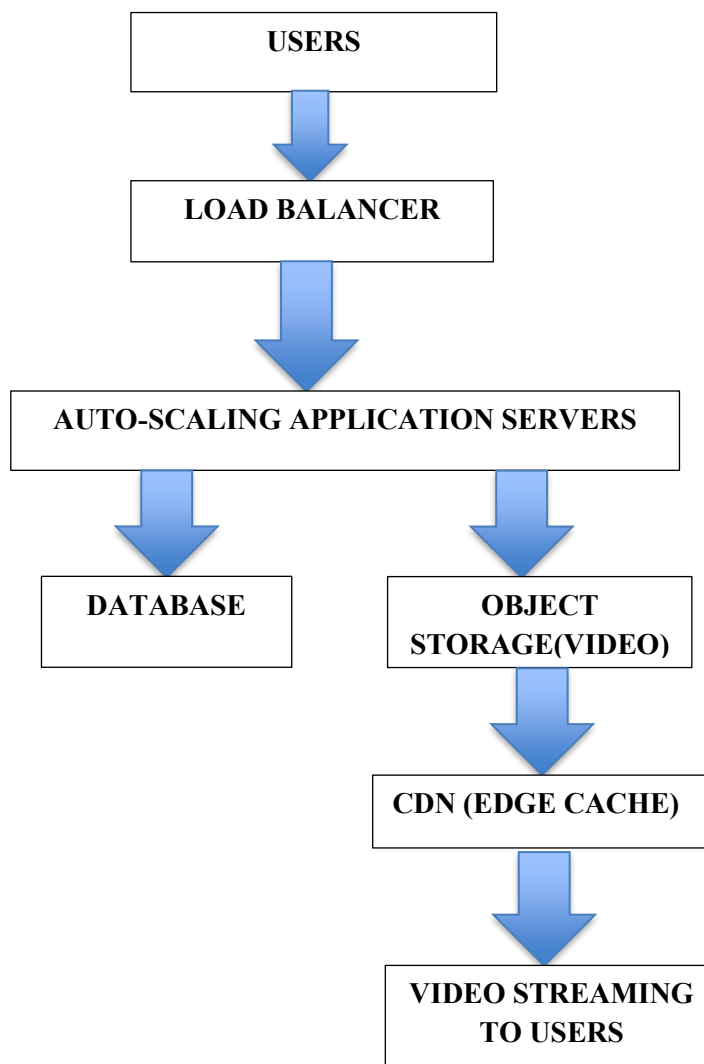
Answer:

1. Cloud Architecture Design

A scalable cloud architecture for a video streaming platform includes:

- **Users** access videos through web/mobile applications.
- **Load Balancer** distributes incoming traffic across multiple application servers.
- **Auto-Scaling Group** automatically adds or removes servers based on user demand.
- **Content Delivery Network (CDN)** caches video files at edge locations close to users, reducing latency.
- **Object Storage** (e.g., cloud storage) stores video content securely.
- **Database** stores user accounts, subscriptions, and watch history.
- **Monitoring Service** continuously tracks CPU, memory, network traffic, and response time.

Architecture Flow:



2. How Load Balancing and Auto-Scaling Ensure Smooth Streaming:

Load Balancing-

- Distributes user requests evenly among multiple servers.
- Prevents server overload.
- Improves availability and response time.
- Automatically redirects traffic if one server fails.

Auto-Scaling-

- Adds new servers during peak traffic.
- Removes unnecessary servers during low traffic.
- Maintains consistent streaming performance.
- Prevents buffering during sudden increases in viewers.

3. Cost Optimization Strategies:

- Use **CDN caching** to reduce bandwidth and server load.
- Store videos in **low-cost cloud object storage**.
- Enable **auto-scaling** so servers run only when required.
- Use **reserved or spot instances** for background processing and video encoding.
- Compress videos using efficient codecs (H.265/AV1) to reduce storage and bandwidth costs.
- Monitor resource usage and remove idle resources.
- Use lifecycle policies to move old videos to cheaper archival storage.

Conclusion:

A cloud-based video streaming platform using **load balancing, auto-scaling, CDN caching, and cloud storage** can efficiently support **millions of concurrent users** with **minimal buffering**. Cost is optimized through **elastic scaling, caching, efficient storage, and resource monitoring**, ensuring high performance and reduced operational expenses.

2)A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

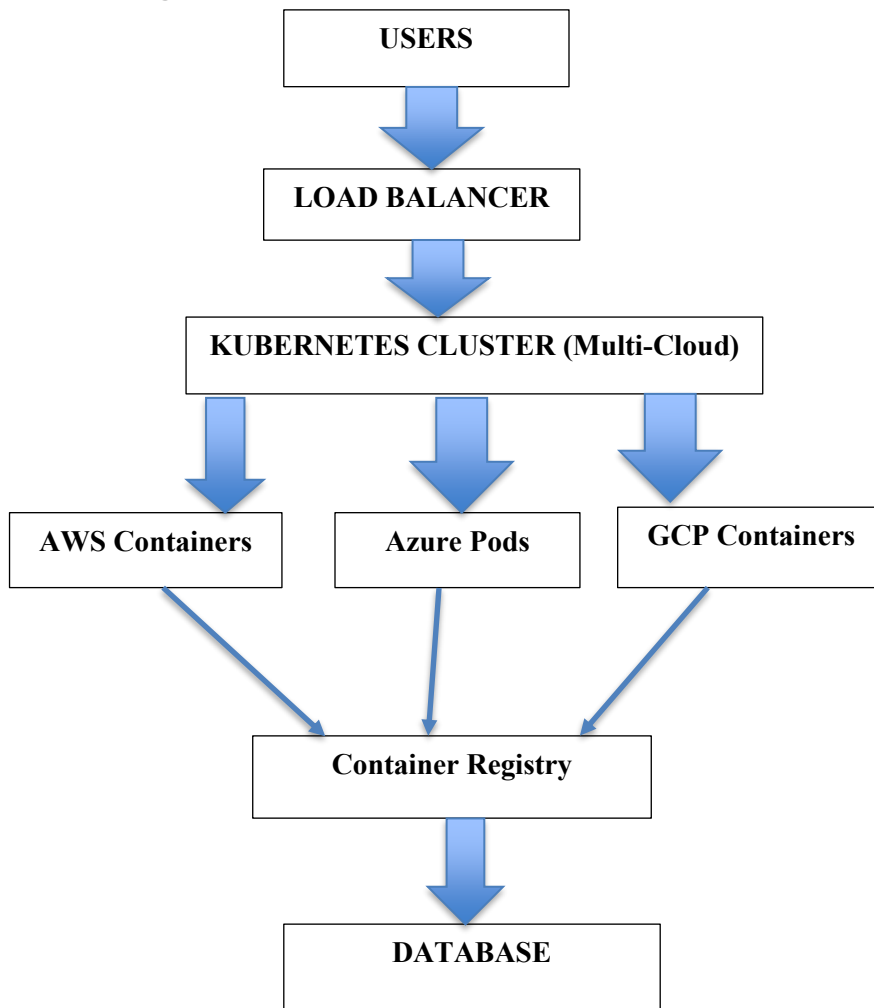
Answer:

1. Proposed Solution

The fintech startup can use Docker for containerization and Kubernetes for container orchestration across multiple cloud providers (such as AWS, Azure, and Google Cloud).

- Package applications using Docker containers.
- Deploy containers on Kubernetes clusters across multiple clouds.
- Use CI/CD pipelines to automate application deployment.
- Store container images in a Container Registry.
- Use a Cloud Load Balancer to distribute user requests.

2. Architecture Diagram



3. Service Discovery and Scaling

Service Discovery

- Kubernetes automatically assigns DNS names to services.
- Containers communicate using internal service names instead of IP addresses.
- Failed containers are automatically replaced.

Scaling

- **Horizontal Pod Autoscaler (HPA)** automatically increases or decreases the number of pods based on CPU or memory usage.
- **Cluster Autoscaler** adds or removes worker nodes when required.
- Load balancer distributes traffic evenly across all running containers.

4. Benefits of Multi-Cloud Deployment

- High availability and fault tolerance.
- Avoids vendor lock-in.
- Better disaster recovery.
- Improved application performance by choosing the nearest cloud region.
- Flexibility to use the best services from different cloud providers.

5. Risks of Multi-Cloud Deployment

- Higher infrastructure and management complexity.
- Increased networking and data transfer costs.
- Security policies must be managed consistently across clouds.
- Monitoring and troubleshooting become more difficult.
- Data synchronization between cloud providers can be challenging.

Conclusion

Using **Docker**, **Kubernetes**, and a **multi-cloud strategy** enables the fintech startup to achieve consistent deployments, automatic service discovery, efficient scaling, and high availability. Although multi-cloud improves reliability and flexibility, it also introduces additional management, security, and cost challenges.

3)An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

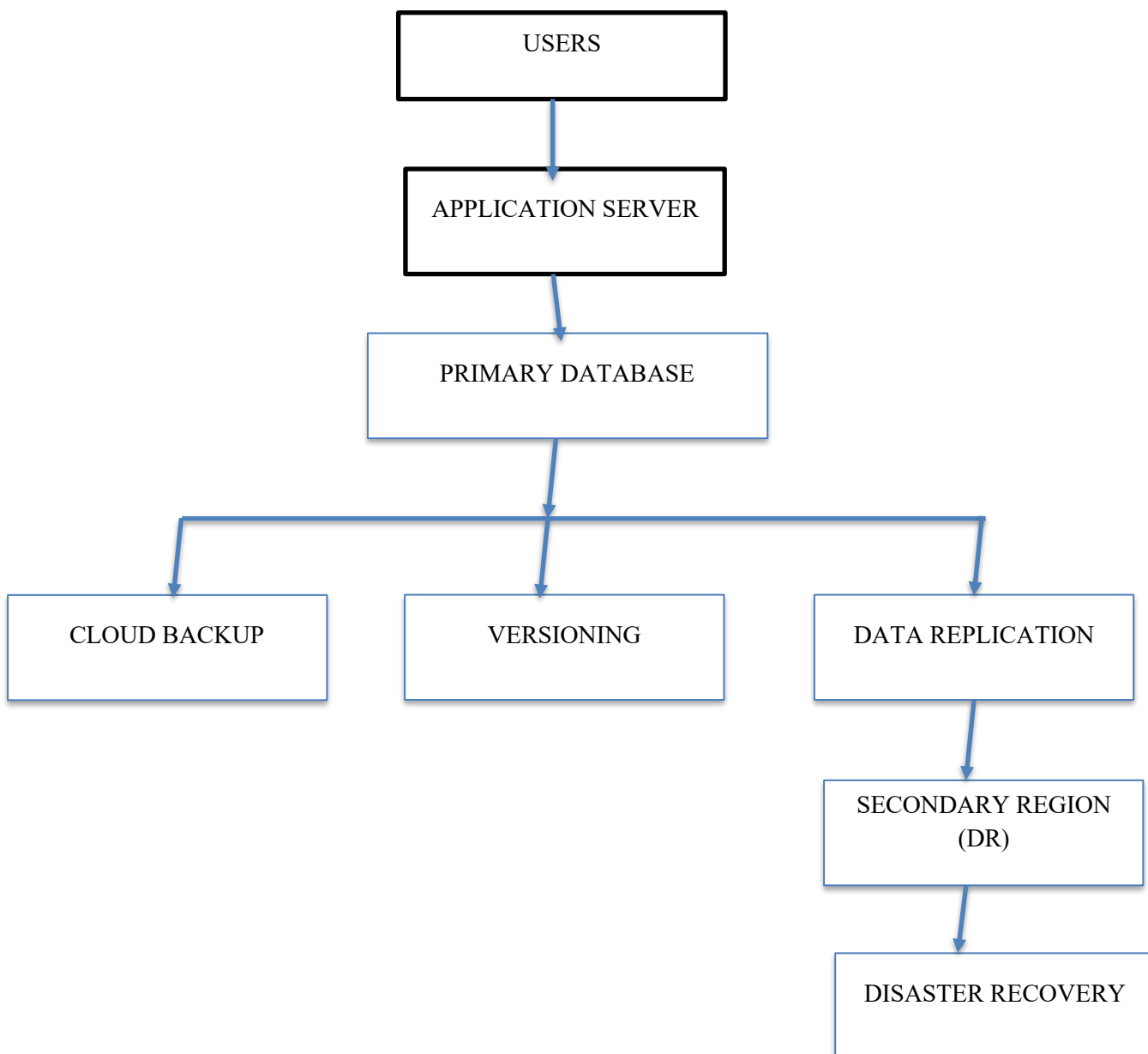
Answer:

1. Solution Design

The e-learning platform should use **cloud-native backup, data replication, and object versioning** to protect data and recover quickly from accidental deletion or system failures.

- Enable automatic cloud backups for databases and files.
- Store backup copies in multiple cloud regions.
- Enable object versioning to recover deleted or modified files.
- Use real-time replication between primary and secondary cloud regions.

2. Architecture Diagram:



3. Achieving RTO and RPO

Recovery Time Objective (RTO):

- **Use automatic failover to the secondary region.**
- **Keep backup servers ready for quick recovery.**
- **Minimize downtime by restoring services within minutes.**

Recovery Point Objective (RPO):

- **Enable continuous data replication.**
- **Perform frequent automatic backups.**
- **Use versioning to restore the latest valid data with minimal data loss.**

4. Steps to Ensure Business Continuity:

- **Schedule automatic daily and incremental backups.**
- **Enable object versioning for all important files.**
- **Replicate data to another cloud region.**
- **Regularly test backup restoration and disaster recovery plans.**
- **Configure automatic failover for critical services.**
- **Continuously monitor backup status and system health.**
- **Maintain documented disaster recovery procedures.**
- **Encrypt backup data and apply access controls for security.**

Conclusion:

By using cloud-native backups, data replication, object versioning, and disaster recovery services, the e-learning platform can recover quickly from accidental deletion, achieve low RTO and RPO, and ensure uninterrupted learning services during failures.

4)A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

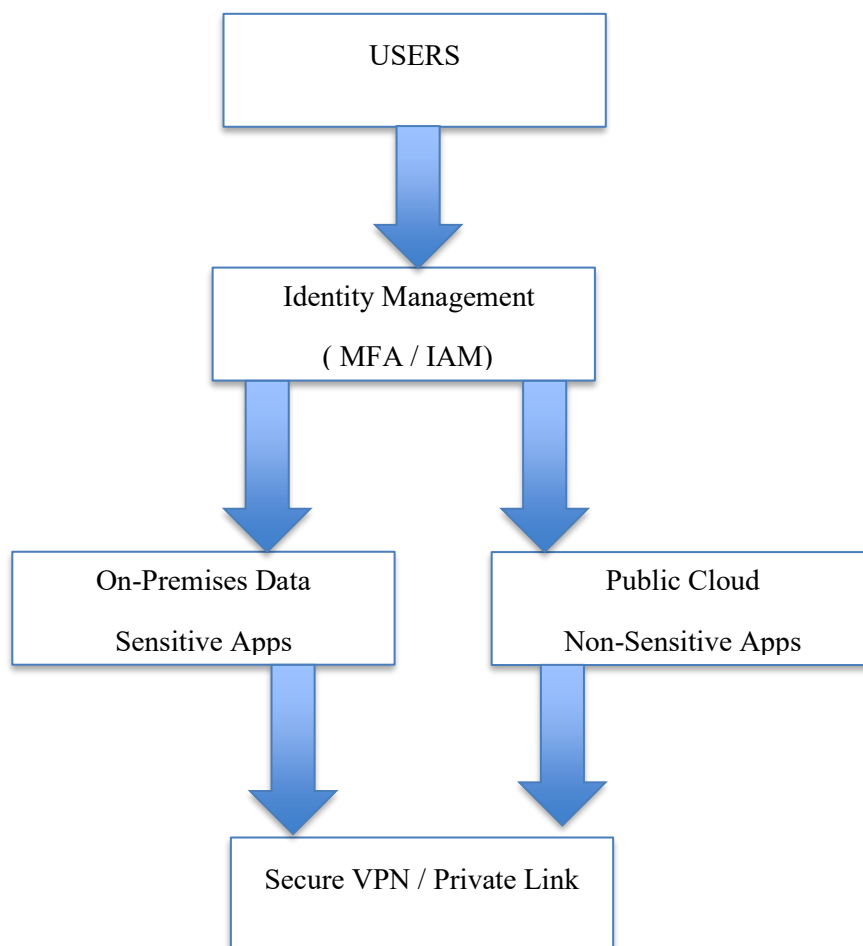
Answer

1. Hybrid Cloud Solution

The government agency can use a Hybrid Cloud Architecture where:

- Sensitive data and critical applications remain On-Premises.
- Non-sensitive applications and public services are deployed in the Public Cloud.
- A VPN or Dedicated Secure Connection securely connects both environments.
- Identity management provides secure authentication across both platforms.

2. Architecture Diagram



3. Identity Management and Encryption

Identity Management

- Implement Identity and Access Management (IAM).
- Enable Single Sign-On (SSO).
- Use Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Store encryption keys securely using a Key Management Service (KMS).

4. Challenges in Hybrid Cloud

- Complex infrastructure management.
- Maintaining consistent security policies.
- Data synchronization between environments.
- Higher networking costs.
- Monitoring multiple platforms.
- Regulatory compliance management.

Conclusion

A hybrid cloud allows sensitive workloads to remain on-premises while leveraging cloud scalability for non-sensitive applications. Secure communication, IAM, and encryption ensure compliance and data protection.

5) A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

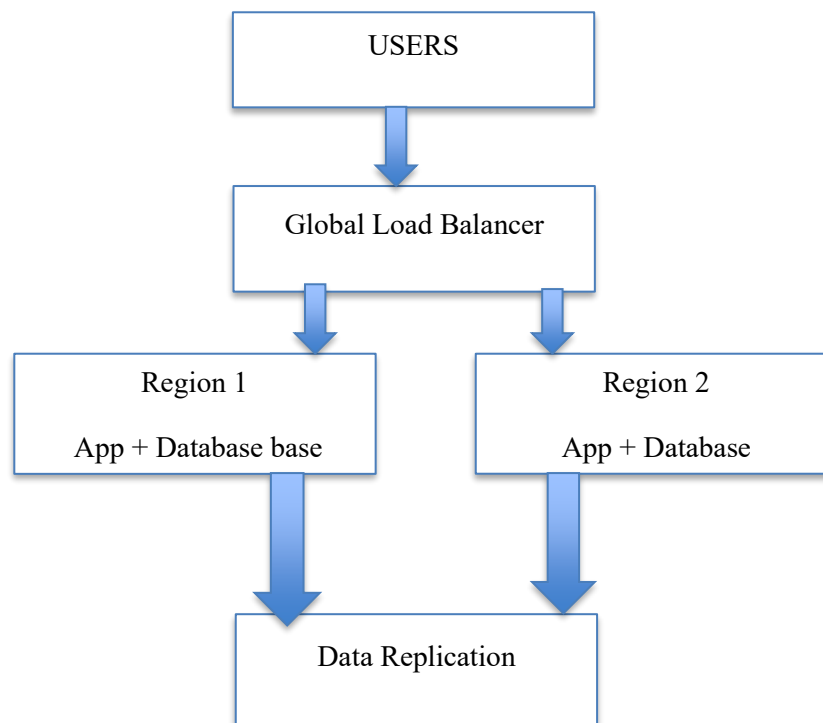
Answer

1. Multi-Region Deployment Strategy

The SaaS provider should deploy the application across **multiple cloud regions**.

- Users connect through a **Global Load Balancer**.
- Traffic is routed to the nearest healthy region.
- Each region hosts identical application servers and databases.
- If one region fails, traffic automatically shifts to another region.

2. Architecture Diagram:



3. Data Replication and Failover

Data Replication

- **Enable continuous database replication between regions.**
- **Maintain synchronized application data.**
- **Use automatic backup and snapshots.**

Failover

- **Configure automatic failover.**
- **Health checks detect regional failures.**
- **Global load balancer redirects users to the healthy region.**
- **Restore failed services after recovery.**

4. Monitoring and Alerting

- **Monitor CPU, memory, disk, and network usage.**
- **Track application response time and availability.**
- **Configure alerts for failures and abnormal resource usage.**
- **Collect centralized logs for troubleshooting.**
- **Perform regular disaster recovery testing.**

Conclusion

A multi-region cloud deployment with data replication, automatic failover, and continuous monitoring ensures high availability, fault tolerance, and uninterrupted service even if an entire cloud region becomes unavailable.